With this knowledge and knowing the speed that the feeders are being operated at, the user can decide if to or not to change the spool before the wire runs out.

HMI Wire Left Function

This machine has been supplied with a facility to report how much wire is left on the wire spools based upon the wire diameter and weight.

The screen that monitors this on the HMI interface is MONOTORING – WIRE LEFT

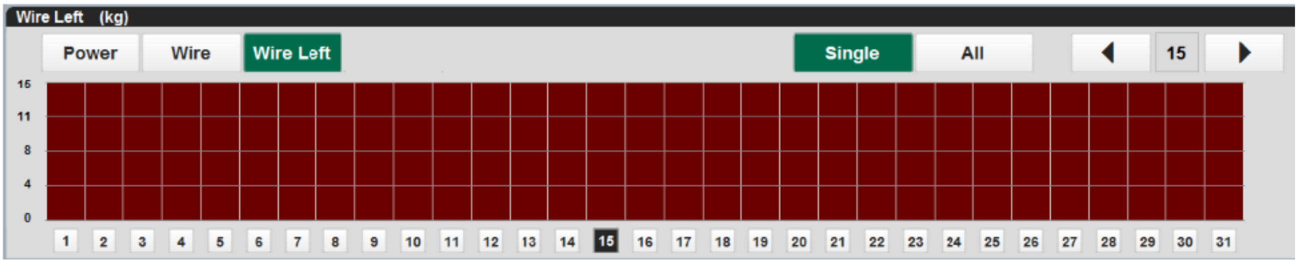


Figure 3.13-3 WIRE LEFT HMI interface screen.

The y scale on the left-hand side of the screen is in KG weight. This will be updated before tension is applied to the substrate.

The weight of aluminium spool and the diameter are entered in the DIAGNOSTICS – MAINTENANCE – CONFIGURE screen.

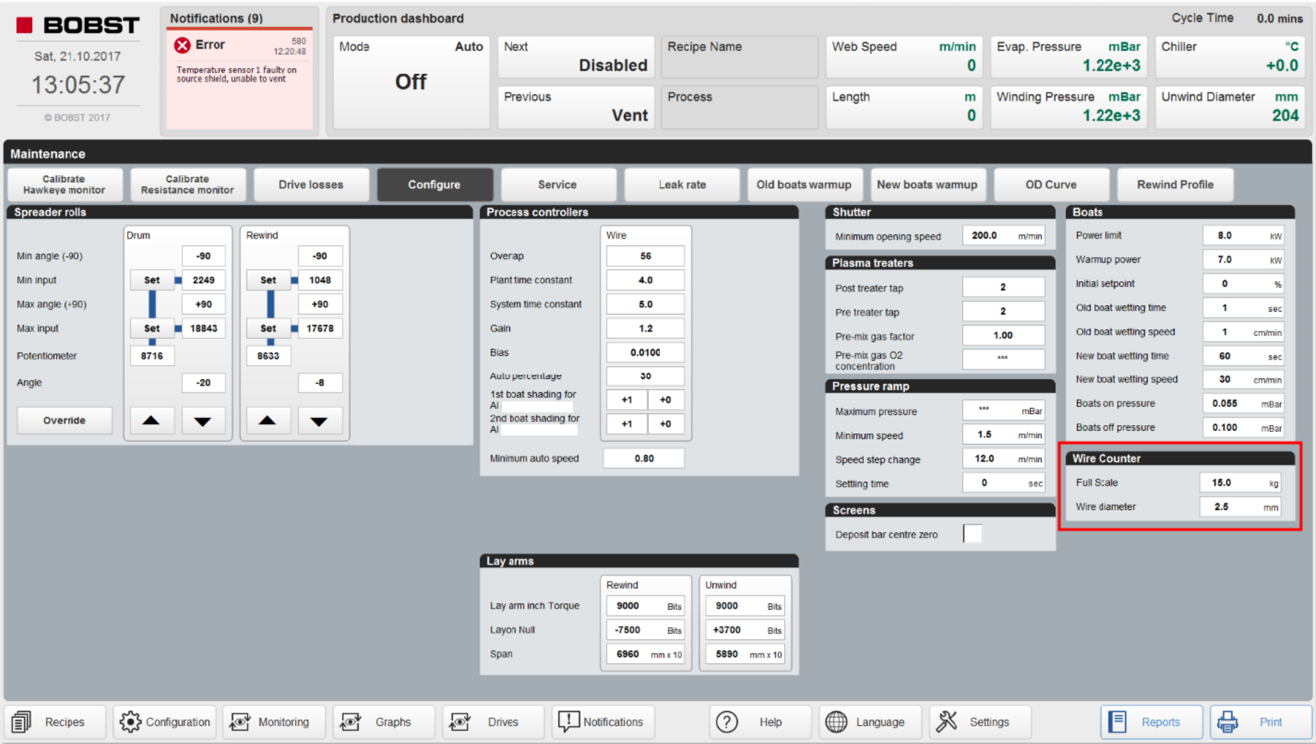


Figure 3.13-4 CONFIGURE HMI interface screen.

Changing the aluminium spools of wire, will require the resetting of this function.

Changing a single spool of aluminium wire

In the case where an individual spool, move the cursor under the spool being replaced and press the RESET WIRE LENTGH button.

Figure 3.13-5 Cursor movement controls

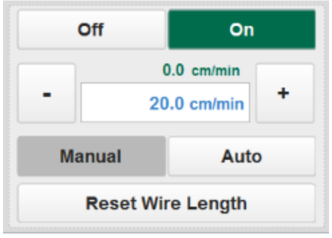


Figure 3.13-6 RESET WIRE LENTGH button

Changing all the aluminium spools

When the ALL button is selected, pressing RESET WIRE LENGTH will reset the spools for the wires that are selected for the current web width. To reset all the wires, select full width first.

3.14 Replacement of copper contact blocks

The copper contact blocks can degrade over time and may need to be replaced.

Different Contact Blocks

The contact blocks installed on a 125mm x 40mm source are shown in the drawing below for identification

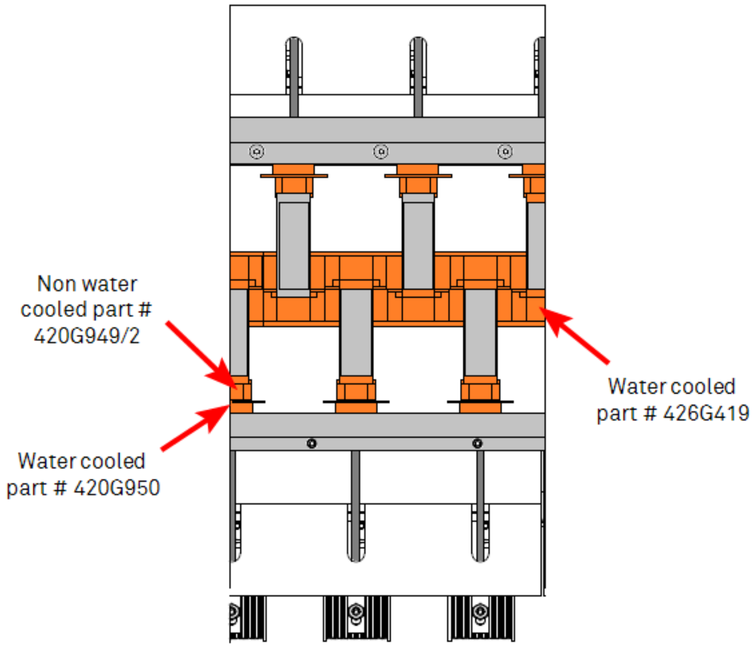


Figure 3.14-1 Identification of evaporation contact blocks

As the contact surfaces between the evaporator and the copper block degrades, the block can be rotated through 180 degrees to make use of the other machined surface.

Isolation of water supply to evaporation area

The machine has been designed to allow the isolation and draining of the water from the source area in the case where blocks are to be replaced or seals be inspected.

	Note !
	Correct water flows in this area are essential and at no time during normal operation, should the water flows be restricted

With the chamber opened and the winding cart fully withdrawn, the main pneumatically operated evaporation source isolation valve on the feed line will be closed. When open, the yellow visual indicator can be seen as shown in the photograph below.

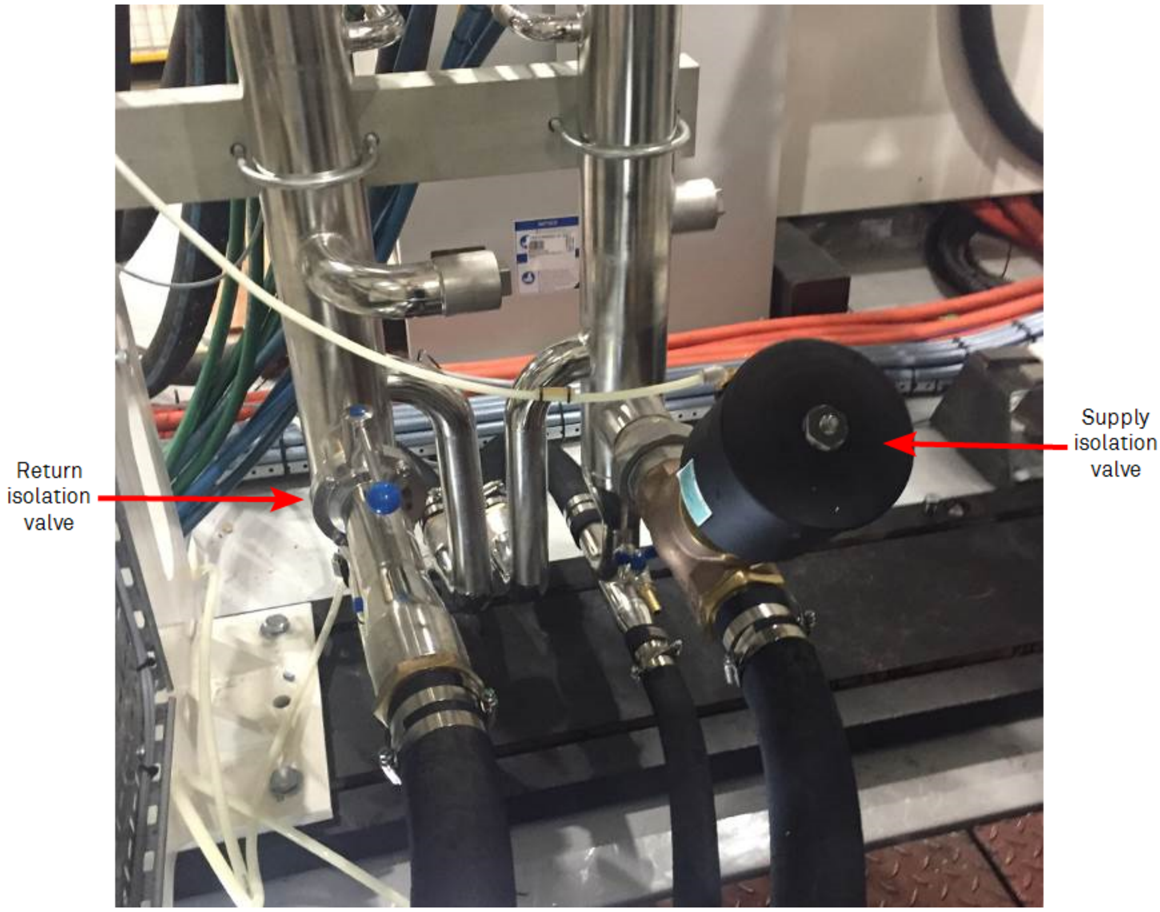


Figure 3.14-2 Evaporation source water supply manifold

Figure 3.14-3 Evaporation source drains

To isolate the complete evaporation area, the following procedure should be followed.

1. Open the chamber and place the winding cart to the fully open position.
2. Close the return isolation valve.
3. Place a suitable container or drain line from the drain valve
4. Open the drain valves.
5. As circuits are broken within the evaporation area, water will flow from the drain lines.

After carrying out the task, the following steps should be taken

1. Close the drain valves.
2. Open the return isolation valve.
3. In the maintenance access level, enable the water flow at the water diagnostic screen by touching the valve icon3. At the HMI maintenance access level, enable the water flow at the DIAGNOSTICS- SERVICES HMI screen by touching the override button.

Figure 3.14-4 DIAGNOSTICS- SERVICES HMI screen

4. Check that the supply isolation valve has opened.
5. Visually check in the area that has been maintained that there are no obvious water leaks.

	Warning !
	Do Not Override Any Of The Machine Interlocks In This Area

Changing the Copper Contact Block – Neutral Side

The copper blocks on the neutral side (non-spring loaded)

Refer to the drawings/photographs below and remove bolts shown below to release the copper contact block.

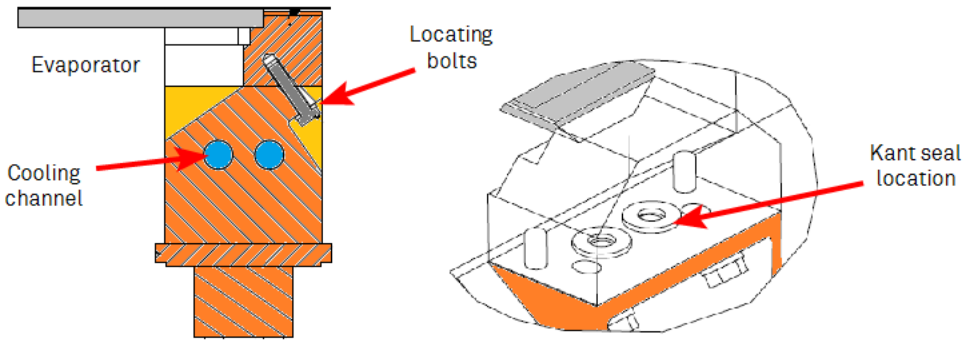


Figure 3.14-5 Neutral copper contact section – Direct water-cooled

After isolating the water, the following procedure should be followed:-

1. Remove the evaporator from between the blocks
2. Clean the aluminium debris from the block and the area close to where it is located.
3. Loosen the two steel bolts.
4. By hand, remove the copper block.
5. If changing the copper block for a new one, replace the Kant seals, at this time.
6. The contact area between the two copper surfaces should be checked and cleaned.
7. The bolt threads should be coated with Rocol J166 anti-seize compound. A data sheet can be found in the consumables section of this manual.
8. A small amount of vacuum grease can be used to hold the Kant seal in the copper block to locate them while re-installing.
9. Ensure that the location pins are located.

	Note !
	Leave the location pins in the evaporation source box wall, do not put them in the copper block, as the holes in the block are deeper that the length of the pin.

10. Locate the block onto the pins
11. Insert the bolts and tighten them to the recommended torque setting.

Changing the Copper Contact Block – Live Side

The blocks are non-cooled and can be changed without removing the water from this area.

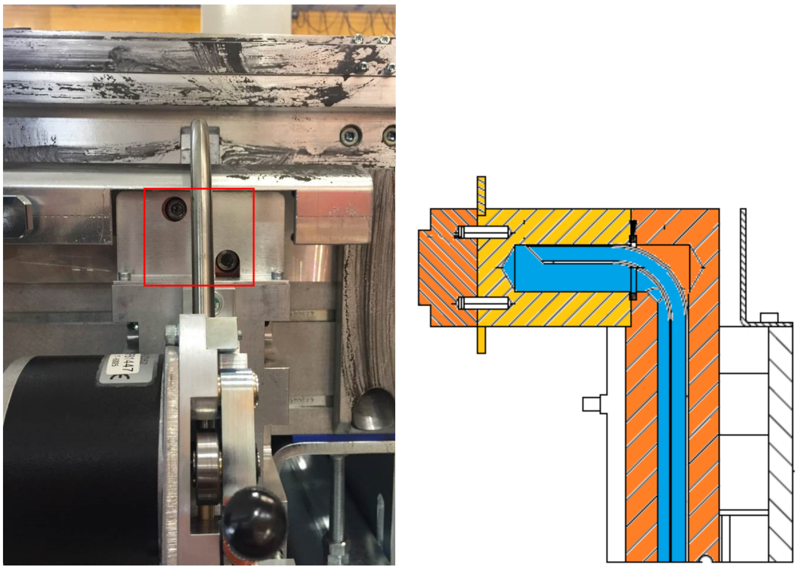


Figure 3.14-6 Copper contact block release bolts

Release the visible cap head bolt to release the non-cooled copper contact block, as viewed above

As the contact surfaces between the evaporator and the copper block degrades, the block can be rotated through 180 degrees to make use of the other machined surface.

The bolt threads should be coated with Rocol J166 anti-seize compound.

The contact area between the two copper surfaces should be checked and cleaned

3.15 Inline Monitor System and Control

There is a process requirement to monitor the amount of aluminium coating being deposited upon the substrate.

Low deposition rates will render the finished product unsuitable, if the coating deposit is not thick and uniform enough to meet the end users requirements.

Heavy deposition rates will allow the process to become non-efficient.

The end user of the substrate will generally request a deposit measured in optical density or Ohms/ Square, between a minimum and maximum level.

Therefore, the uniformity across the width of the substrate and down its processed length is very important.

An inline measurement system is provided to give feedback to the operator and the machine control system, which can make adjustments, as required.

This monitor system uses sensors situated near to the film surface. These are located in a beam mounted in the winding mechanism.

This K5 Expert is equipped with the following monitor beams:

- 5 Probe Optical Monitor Beam, with probes located so that they align with the 'control lanes' in a structured product.

The sensors send signals to the computer display on the PROCESS screen. This displays the deposition coating value, numerically and on a bar chart.

There are a number of displays that can be seen on the MONITORING HMI screen, including:

- Deposition levels in a bar chart format on the machine HMI screen
- Wire feed rates in a bar chart format.
- Power levels supplied to the evaporators.

The deposition can be displayed in three standard formats:

- Light Transmission (%T)
- Resistance (Ohms/ square)
- Optical Density (OD)

Details about each is described later in this section.

Figure 3.15-1 Screen capture of the HMI PROCESS screen

The screen allows the user to monitor the deposition being coated on to the substrate.

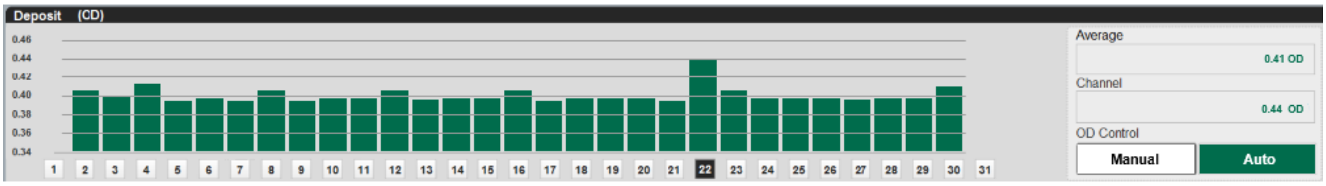


Figure 3.15-2 Deposition section of the MONITORING screen

The parameters on the right-hand side of the screen, give the average value across all the sensors being used, and the value of the sensor that cursor is located under

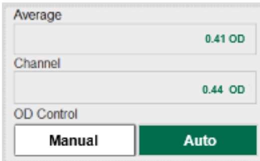


Figure 3.15-3 Deposition section of the MONITORING screen

Selection of Display Mode

The selection of the display mode is made at the time of setting up the recipe

Touch the drop down tab on the product target option to list the types of display units.

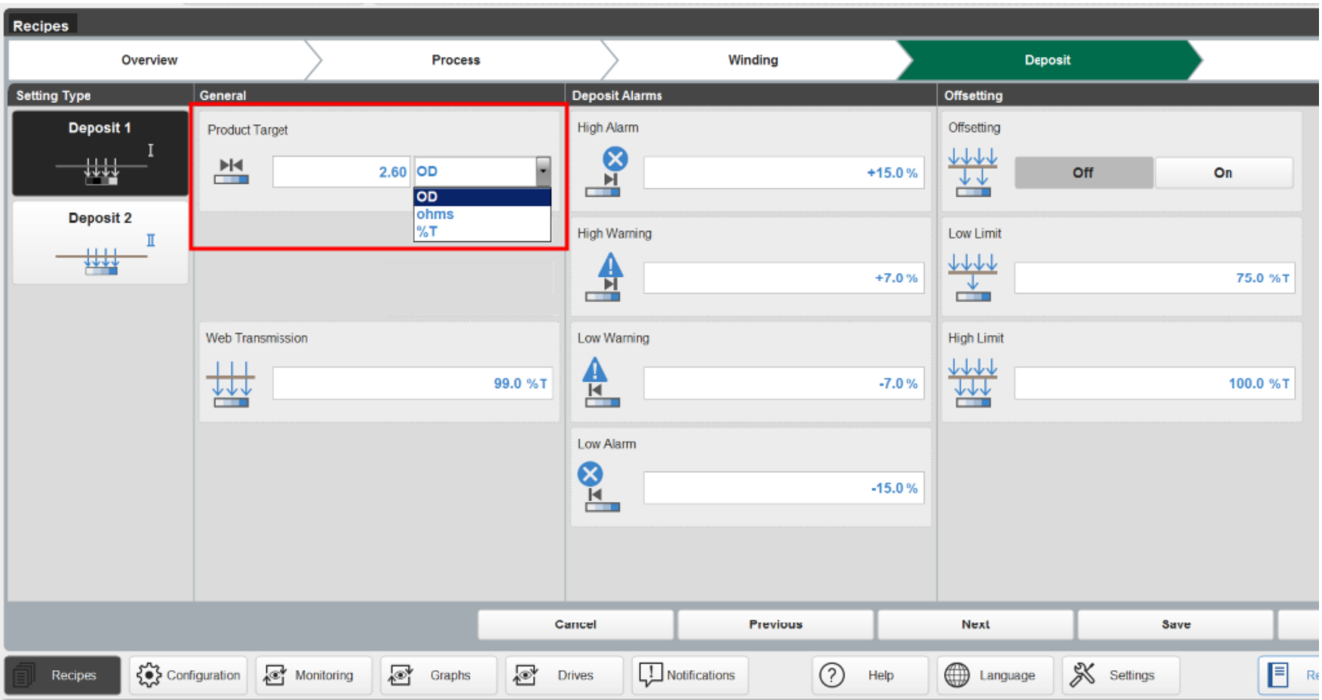


Figure 3.15-4 Deposition section of the RECIPIES- DEPOSIT- HMI screen

Resistivity Display

The amount of coating measured by the monitoring system can be displayed as a resistance (Ohms/square) reading.

The display will show the bars at the top of the screen when there is no aluminium being coated onto the web. As the aluminium is started to be evaporated onto the web, the display will change and the bars will reduce in height.

If the coating is thick, the bars will be at the bottom of the screen.

The target of the deposition previously requested is displayed in the left-hand side of the display..

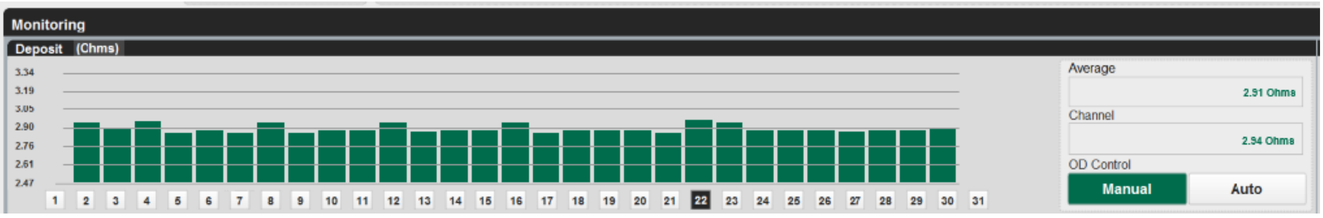


Figure 3.15-5 Deposition section of the process screen – resistance display

Optical Density Display

The display will show the bars at the bottom of the screen when there is no aluminium being coated onto the web.

As the coating is started to be evaporated onto the web, the display will change and the bars will increase in height.

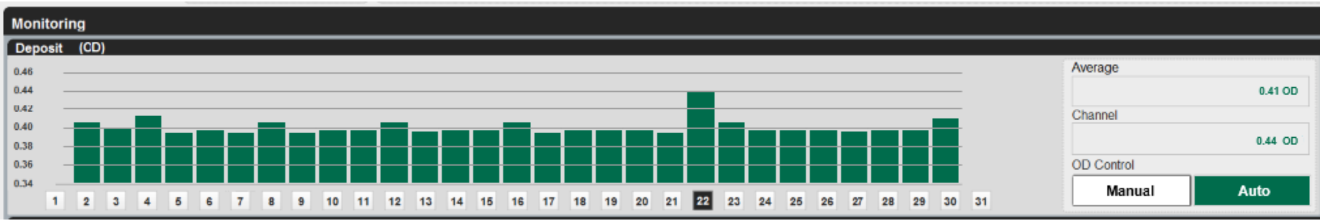


Figure 3.15-6 Deposition section of the MONITORING HMI screen - optical density.

Light Transmission Display

The display will show the bars at the top of the screen when there is no aluminium being coated onto the web.

As the coating is started to be evaporated onto the web, the display will change and the bars will decrease in height.

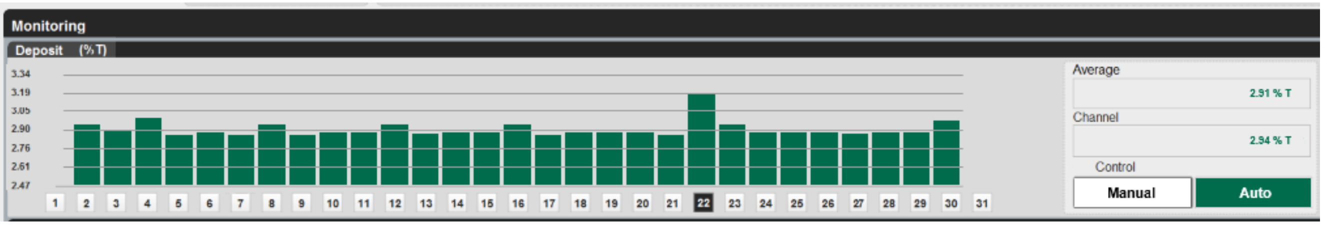


Figure 3.15-7 Deposition section of the MONITORING HMI screen – light transmission

Optical Density v Resistivity Curve

The base optical density of individual substrates is different, for example CPP in comparison with PET.

It has been also found that for the same aluminium thickness, as measured in resistance, the optical deposition/transmission can be different between substrates.

Therefore individual optical density v resistance curves can be created for each RECIPE

There is a conversion curve (OD) in the RECIPE screen to allow for the conversion of the Hawkeye produced optical density readings to resistance readings. These curves are recipe based.

This function is only available at the higher login access level.

The button should be touched and then the curve button.

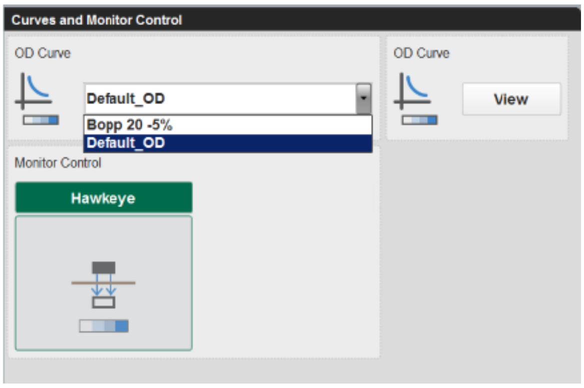


Figure 3.15-8 OD Curve selection buttons

If an existing curve is to be used in the menu, the OD curve selection drop down button on the left hand side, should be touched to display previously saved curves.

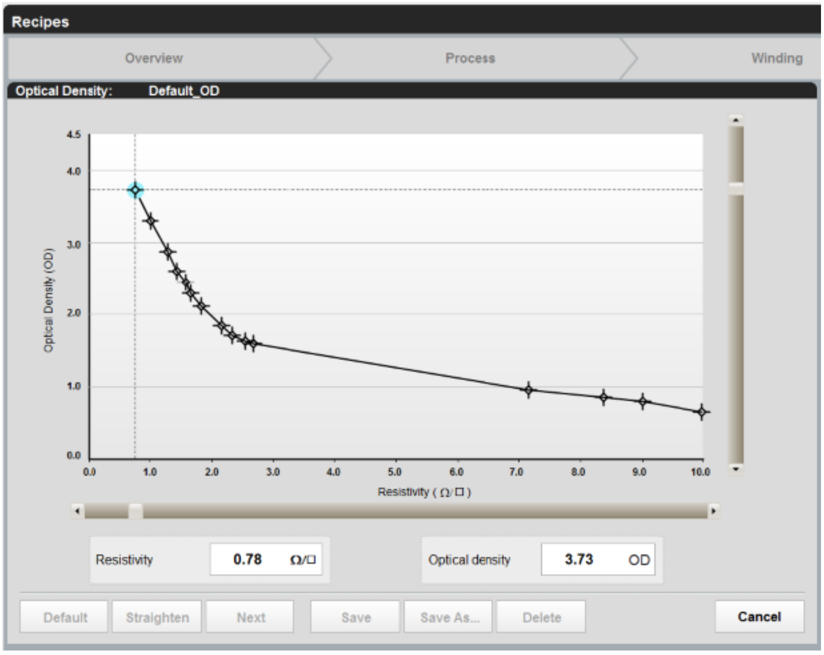


Figure 3.15-9 Optical density V Resistance curve

The ohms and optical density values for each of the cross hairs are displayed at the bottom of the screen.

The current values shown are for the “blue” highlighted “crosshairs” point.

The curve is plotted by touching the screen on any of the points and modifying either the Resistivity or the optical density value. When these are modified, the point will move to the cross reference point.

Button	Purpose/explanation
Default	Selection of this button will select a Bobst created default curve.
Straighten	Selection of this button can be used to create a curve between two points. For example selecting the first and last points on the curve and pressing this button will create a straight line between them with the other crosshairs spread down its length.
Next	If a point is highlighted, pressing this button will move the blue highlight to the next point. This occurs from left to right.

Table 3.15-1 Curve editor buttons - Explanation

Button	Purpose/explanation
Save	Selection of this button will save any modifications to the opened curve.
Save As...	Selection of this button will open up a further dialogue box as described later, to allow the curve to be saved under a different name
Delete	Selection of this button will allow the user to delete the current curve
Cancel	Selection of this button closes the screen

Table 3.15-1 Curve editor buttons - Explanation

Save As Function

As an example the steps below show a curve called “test” being created.

Select the SAVE As button and the HMI asks the user to name the new curve. In this case we call it “test”

Figure 3.15-10 Example of creating a “saved” OD curve

Delete Function

As an example the steps below show the curve called “test” being deleted.

After selecting the curve from open drop down list, and it being displayed, press the YES button.

A pop up screen will be shown asking the user to confirm this action.

Figure 3.15-11 Example of deleting an OD curve

Online v Laboratory Readings

To allow for differences between online optical density readings and offline densitometers to read the same value while metallizing of aluminium to take place, a conversion table has been provided.

The following companies typically make offline devices.

- Tobias
- Macbeth
- Optel
- X-rite
  - The X-Rite 301 Densitometer provides highly repeatable and accurate measurements of black and white x-ray film densities:

Figure 3.15-12 X-Rite 301 transmission densitometer

There is a requirement for this conversion, as the light source (wavelength) that the aluminium deposit is measured at generally varies from offline laboratory instruments, used by customers and the light source used in the Hawkeye system.

The typical scenario is that the customer reports that their laboratory claims that the target deposition on the machine is read lower on the laboratory offline equipment. This requires the optical density (OD) to be set at a higher level in the metallizer to achieve this offline value.

As the metallizer is also used to produce data reports of the process runs from the Hawkeye reporting computer, and it is not acceptable that the deposition set point and feedback readings are higher than what the final product actually is.

Automatic Deposition Control

The machine HMI system has the following modes for controlling the evaporation of the aluminium deposition.

Stop Mode

Stop Mode is the normal mode at the startp-up of the machine. No control signals are sent to the wire feeder stepper motors and evaporator power supply thyristor units. The evaporator does not heat up, and no wire is fed to the evaporators.

Touching the Boat Power On" button on the MONITORING screen will energise the power supply to the evaporators, via the thyristors.This can be set to operate automatically and is discussed in another section of this manual.

This action also triggers the computer to start the ramp up of the control signal to the thyristors.

The POWER bar chart will be displayed at the bottom of the screen indicating the pre-set evaporator power settings.

The individual power level settings are set by moving the relevant diamond shapes upwards.

These indicators are set between 0 and 100% of the output of the power source.

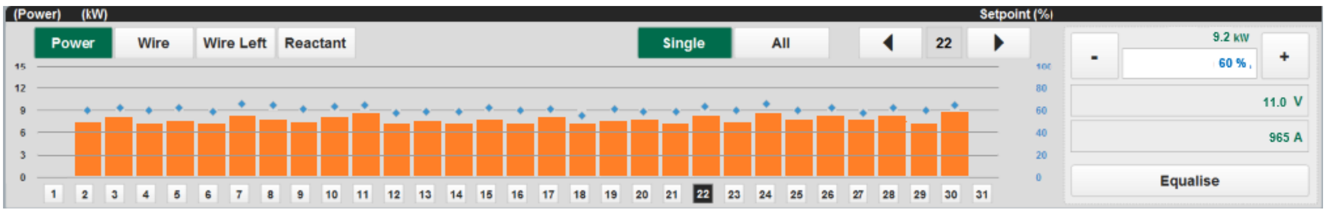


Figure 3.15-13 POWER section of the HMI PROCESS screen

Warm-Up Mode (Standby)

This mode is normally used to apply a signal to the steppers to give a wire feed rate of 100 percent of the nominal base speed from each wire feeder.

For example, if the wire feed rate was set up for 20 cm/min (as displayed on PROCESS screen) and the output bars are at 100%, the actual wire feed rate would be 20 cm from each feeder.

The computer will place the wire feed outputs in this mode once the evaporators have reached their operating temperature. This occurs at the end of the evaporator ramp cycle.

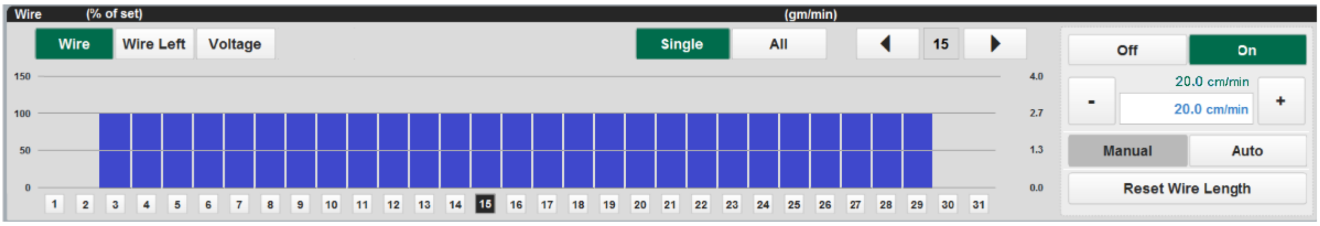


Figure 3.15-14 Wire feed output section of the HMI PROCESS screen.

The wire base speed is set up in the RECIPE screen and can be modified on the MONITORING HMI screen.

If ALL is selected, the + and - buttons adjust the base speed. If ALL is not selected, the + and - buttons adjust the selected wire.

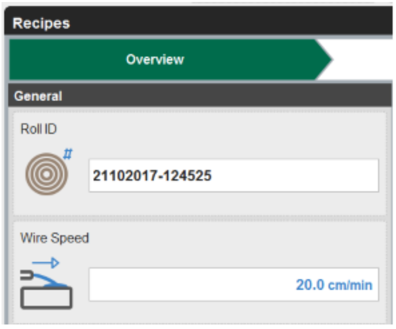


Figure 3.15-15 Wire feeder Set point input on RECIPE OVERVIEW screen

All the wire speed feed set points can be adjusted at once by touching the SINGLE /ALL section of the MONITORING HMI screen. The base wire feed can be adjusted by touching the raise or lower buttons by the side of the wire feed display.

Optimum Set-Point

The software has the ability to pre-set the output levels to give optimum coating.

This is based upon the output values at the end of the previous coating cycle. The warm up values will be set to the same output values. This only occurs if the AUTO mode was selected.

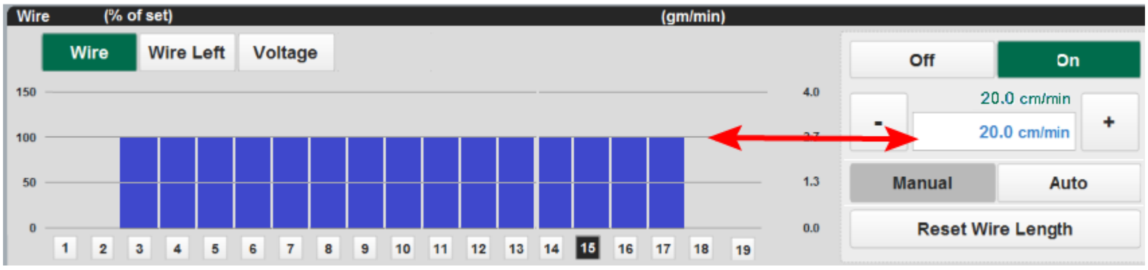


Figure 3.15-16 Wire feed output – optimum set point

100 on this chart equals the base wire feed rate in cm. If set to 20cm/min. this is equivalent to 100.

Resetting Wire Feed Rates to 100%

To reset the warm up values to a 100% output, a RESET button is provided. This can be used when new evaporators are to be supplied. This function can be used in auto or manual mode.

Manual Control Mode

If the user wants to alter the wire feed outputs while the coating process is taking place, it can be done by selecting an individual channel using the cursor. The MANUAL button should be pressed and a “hand icon” will be shown over the channel selected for manual control.

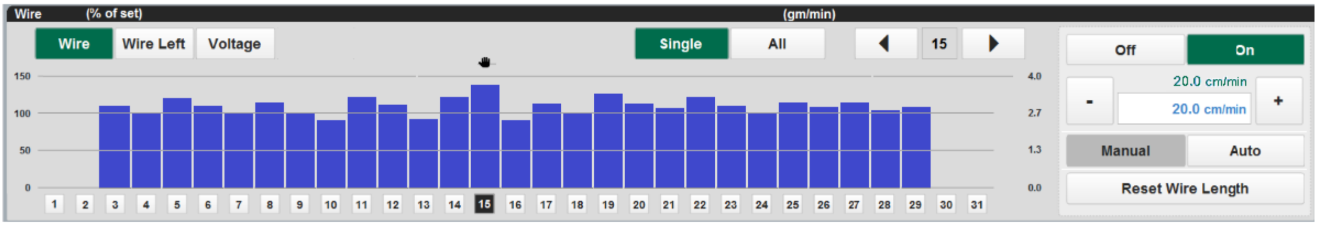


Figure 3.15-17 Wire feed output –single channel manual control

Using the +/- buttons will increase/decrease the wire for the single highlighted channel.

Note !

To deselect the manual operation, the AUTO button should be pressed, and the machine will again take control of the channel

Automatic Control Mode

This mode of operation allows the machine PLC CPU to change automatically from standby mode (STAND-BY) to the control (AUTO) mode, once all of the monitored readings are within 60 % of the preset uniformity value required.

Normally, the user would select this mode once the actual line speed has matched the set speed.

Figure 3.15-18 HMI MANUAL/AUTO control selection button

The mode is selected by touching the CONTROL AUTO/MAN switch on the PROCESS screen.

Once the AUTO mode has been selected, the controller will adjust the wire feed rates via the steppers to give minimal error between the actual and required deposition target.

For any reason that the readings go outside the preset 30% band, the controller will switch back into the Warm up (Standby) mode.

The deposition may move outside the 30% region for a number of reasons and reference should be made to relevant section of this manual for process related problem information.

Also at the end of coating period, the controller will remember and store the output values that were being sent to the wire feeders. This ensures better uniformity earlier into the coating run of the next roll of material.

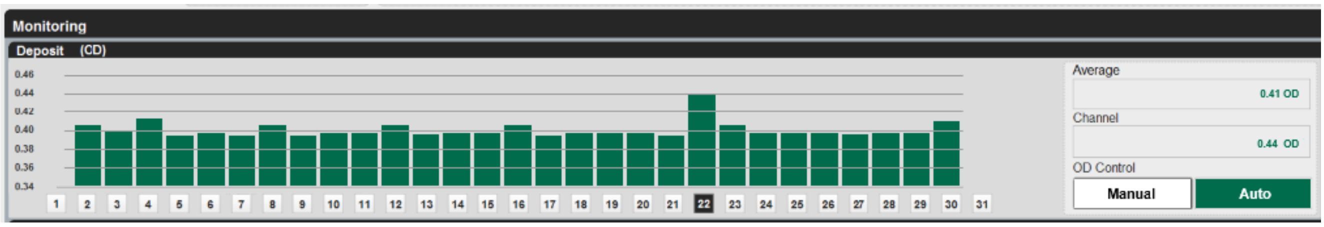


Figure 3.15-19 HMI deposition control display in AUTO mode

The above screen shows the probes in operation and the deposition rates measured on each of them.

Tolerances are set for report purposes, and these are set in the RECIPES HMI screen, as a percentage of the deposition set point.

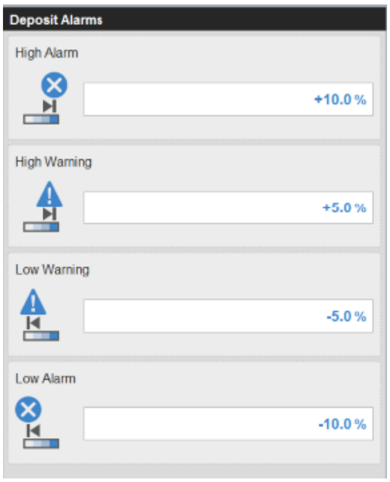


Figure 3.15-20 Deposition system alarm and warning level selection

Note !

The middle of the screen indicates zero error between the actual and set deposition levels.

Moving the cursor below each of the channels can monitor the individual deposition readings. The measurement is indicated in the box located in the below the channel selected.

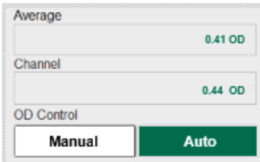


Figure 3.15-21 Monitoring individual channel and average readings

3.16 Adjusting the Evaporation Source Height

This machine has been designed to allow the evaporator to drum height to be varied.

This is accomplished by releasing the location bolts and adjusting the jacking bolts to raise or lower the source. These bolts are duplicated at either end of the source.

An accurate level spirit level should be used to check the alignment in both axis to ensure that the assembly is horizontal.

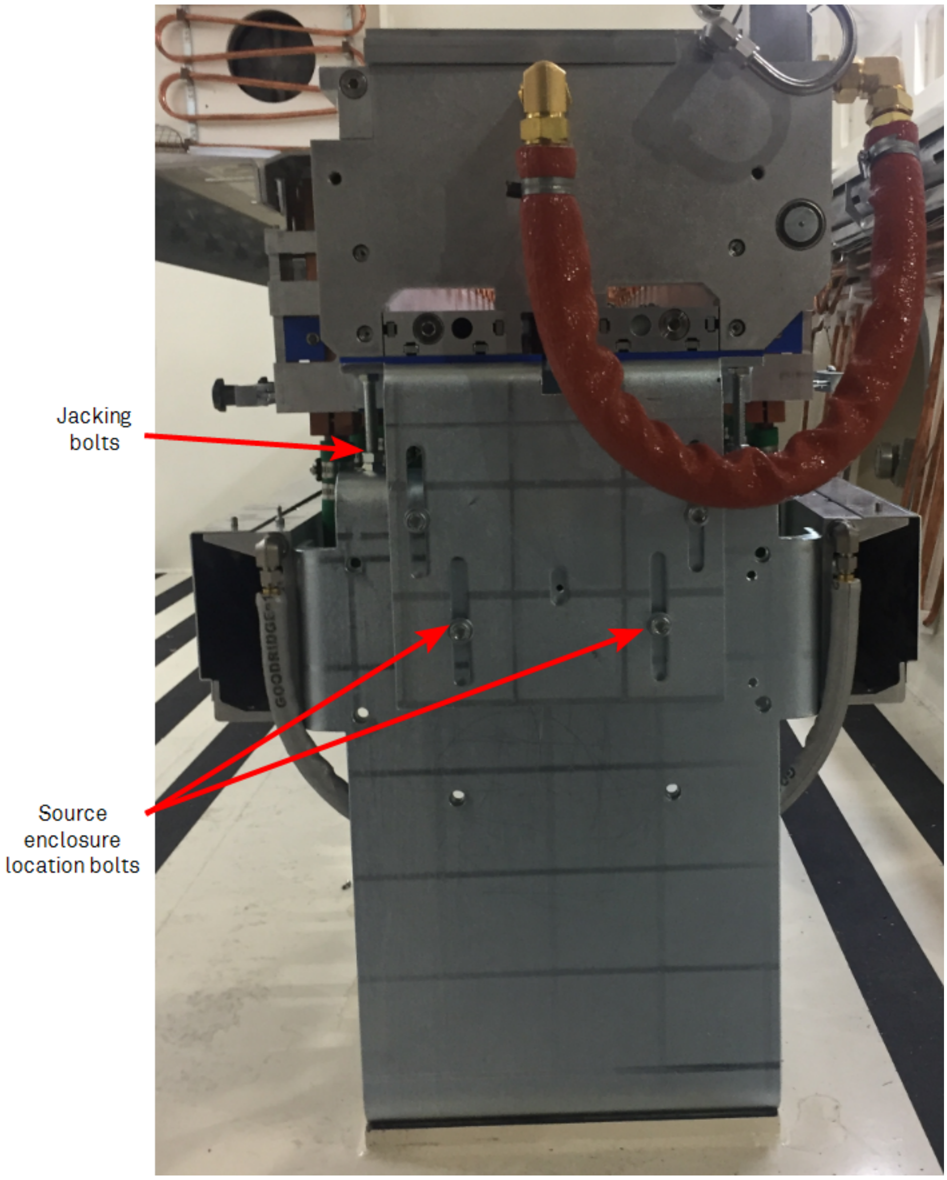


Figure 3.16-1 Source height adjustment points

The closer the evaporators are to the drum, the more heat is transferred onto the web, and the collection efficiency is better than when it is lowered.

Note !

Ensure that both sides of the source are raised evenly. Use a spirit level to check for correct levels.

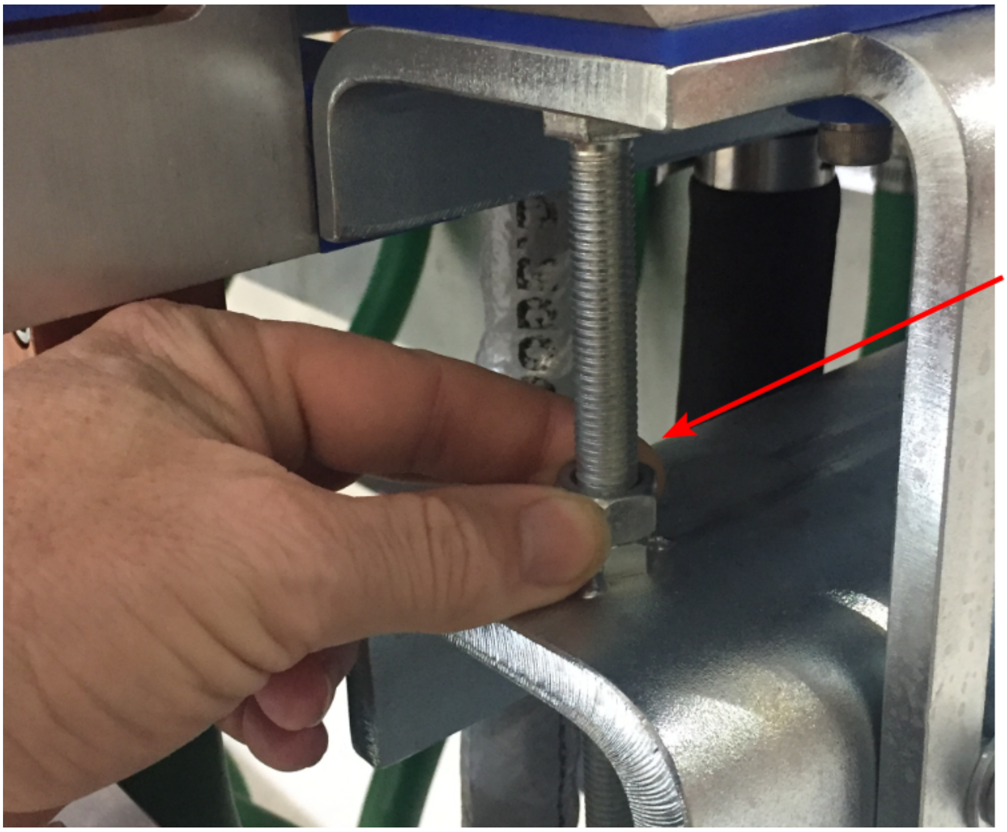
On wider machines there is a central support that has to be adjusted.

Adjusting the Height

The design of the machine allows for adjustment within the limits of surrounding components.

The design allows the upper of the two parallel support brackets to be moved up or down, while the lower one is fixed.

Figure 3.16-2 Source height jacking arrangements



Jacking bolt

Figure 3.16-3 Source height jacking arrangements

	Caution!
	After making adjustments, visually check that there are no potential short circuits on the live post area or restriction of water supply lines

SECTION 4.0 OPERATION OF THE MACHINE WINDING PROCESS

The winding system's function is to transport material from the unwind roll to the rewind roll and allowing the coating processes to take place. The system is designed to process the material without causing any deformations. If faults are present in the parent base roll, it must be recognised that the winding mechanism cannot remove all these defects.